

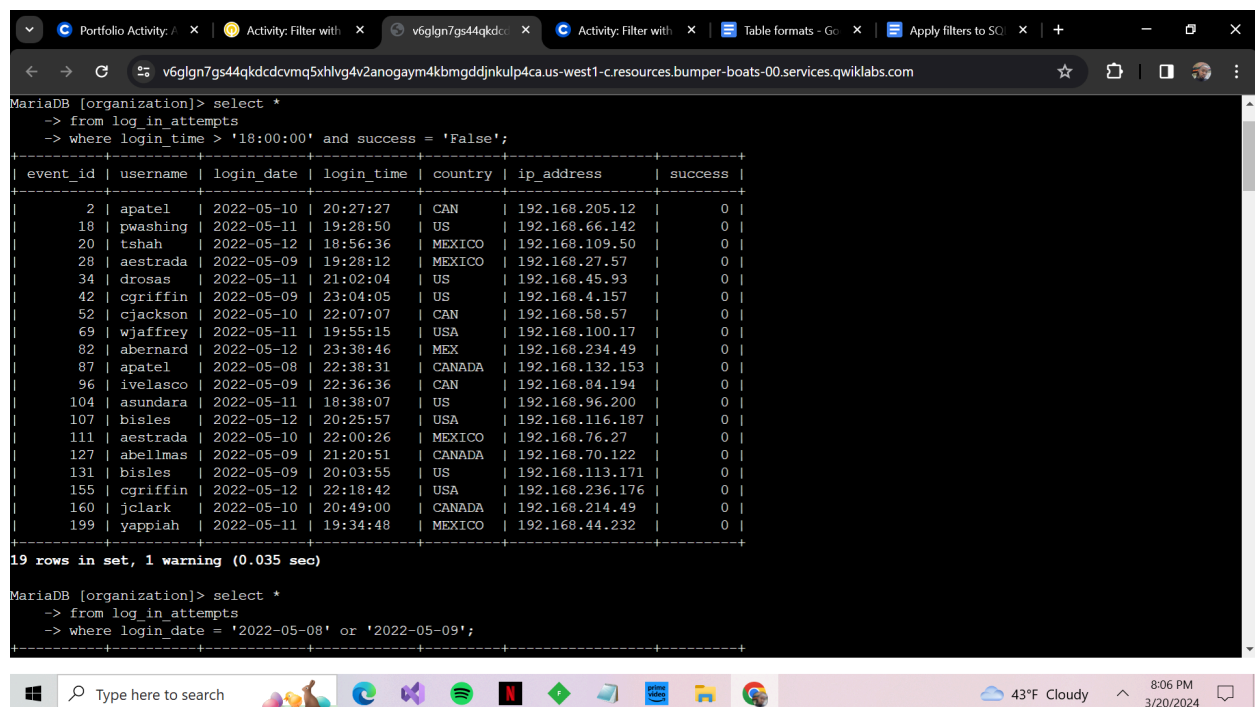
Apply filters to SQL queries

Project description

[I will be accomplishing 6 different tasks that involve filtering with SQL that deal with real world possible scenarios in Cybersecurity. The filters I will work with are where, and, or, like, and not.]

Retrieve after hours failed login attempts

[



```
MariaDB [organization]> select *
-> from log_in_attempts
-> where login_time > '18:00:00' and success = 'False';
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
52	cjackson	2022-05-10	22:07:07	CAN	192.168.58.57	0
69	wjaffrey	2022-05-11	19:55:15	USA	192.168.100.17	0
82	abernard	2022-05-12	23:38:46	MEX	192.168.234.49	0
87	apatel	2022-05-08	22:38:31	CANADA	192.168.132.153	0
96	ivelasco	2022-05-09	22:36:36	CAN	192.168.84.194	0
104	asundara	2022-05-11	18:38:07	US	192.168.96.200	0
107	bisles	2022-05-12	20:25:57	USA	192.168.116.187	0
111	aestrada	2022-05-10	22:00:26	MEXICO	192.168.76.27	0
127	abellmas	2022-05-09	21:20:51	CANADA	192.168.70.122	0
131	bisles	2022-05-09	20:03:55	US	192.168.113.171	0
155	cgriffin	2022-05-12	22:18:42	USA	192.168.236.176	0
160	jclark	2022-05-10	20:49:00	CANADA	192.168.214.49	0
199	yappiah	2022-05-11	19:34:48	MEXICO	192.168.44.232	0

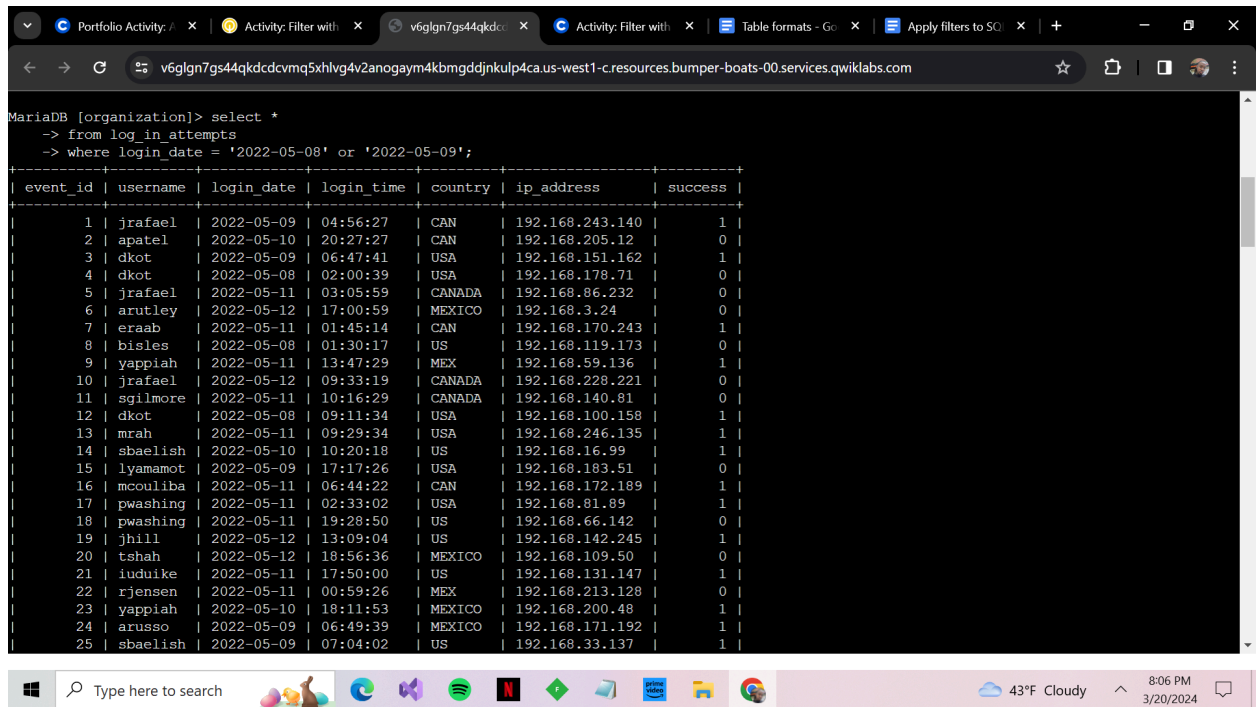
```
19 rows in set, 1 warning (0.035 sec)

MariaDB [organization]> select *
-> from log_in_attempts
-> where login_date = '2022-05-08' or '2022-05-09';
```

The screenshot you see filtering of two separate columns with the where filter along with and. I was able to not only filter for time of login attempts but also combine failed attempts to see any suspicious activity.]

Retrieve login attempts on specific dates

[



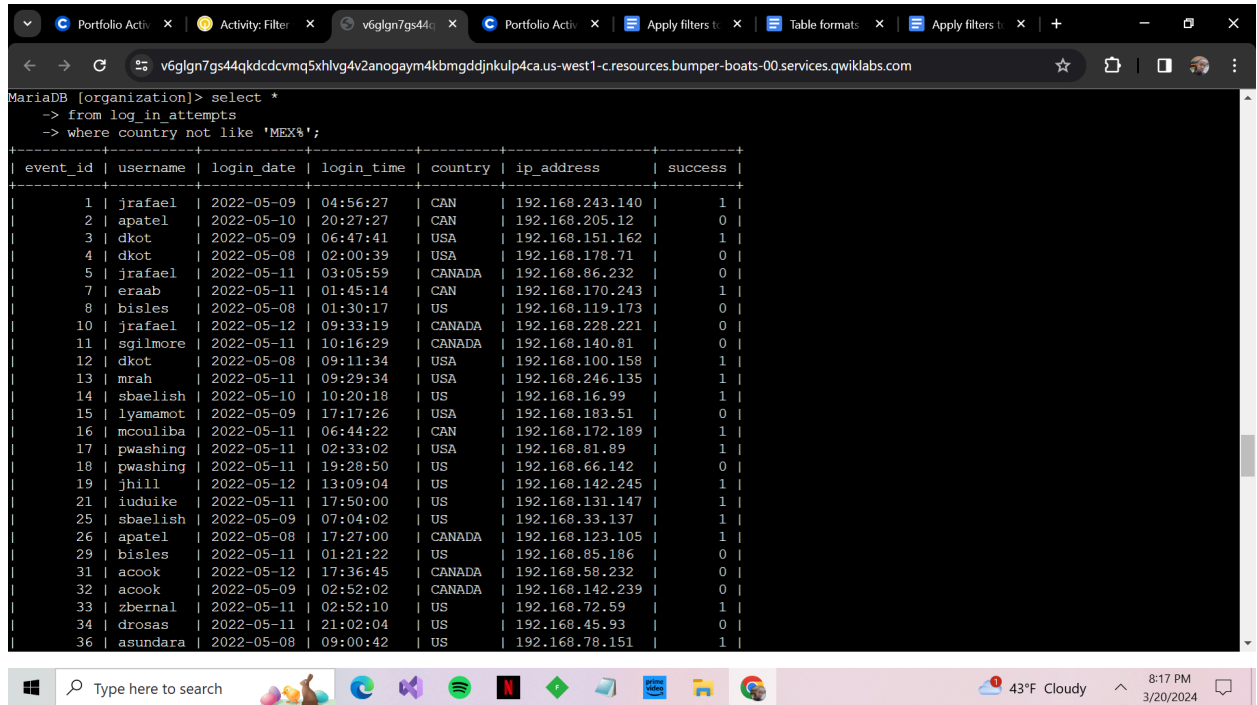
The screenshot shows a web browser window with a MariaDB interface. The browser's address bar displays a URL from qwiklabs.com. The SQL query entered is: `select * from log_in_attempts where login_date = '2022-05-08' or '2022-05-09';`. The results are displayed as a table with 7 columns: event_id, username, login_date, login_time, country, ip_address, and success. The table contains 25 rows of data, showing login attempts for various users on the specified dates. The Windows taskbar at the bottom shows the time as 8:06 PM on 3/20/2024.

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
6	arutley	2022-05-12	17:00:59	MEXICO	192.168.3.24	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
9	yappiah	2022-05-11	13:47:29	MEX	192.168.59.136	1
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrhah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
22	rjensen	2022-05-11	00:59:26	MEX	192.168.213.128	0
23	yappiah	2022-05-10	18:11:53	MEXICO	192.168.200.48	1
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1

The screenshot shows how I was able to retrieve two different dates to be filtered as I used where, and or filters. That allowed me to see any suspicious activity that was alerted on those two days.]

Retrieve login attempts outside of Mexico

[



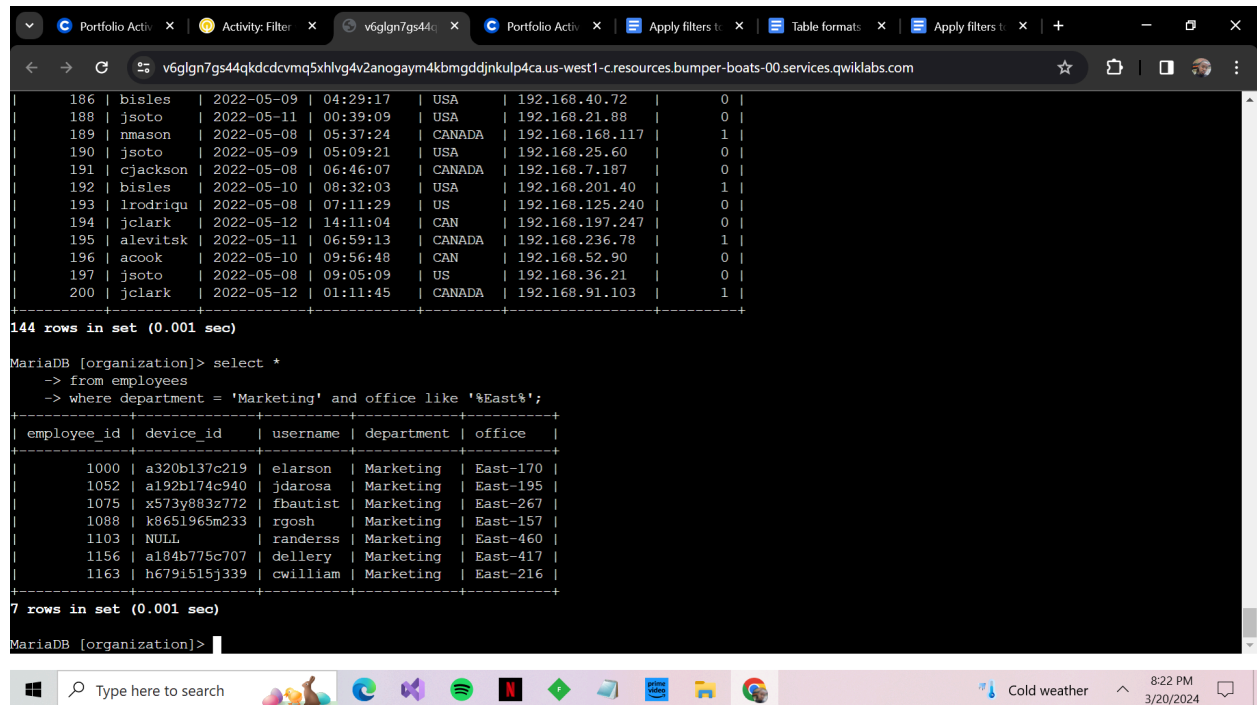
```
MariaDB [organization]> select *
-> from log_in_attempts
-> where country not like 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgillmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1

The screenshot shows how I was able to filter for every state except Mexico by using the where and not filter. This allowed me to check for suspicious activity in other countries that has popped up in the alert.]

Retrieve employees in Marketing

[



The screenshot shows a web browser window with a table of employee data. The table has columns for employee_id, device_id, username, department, and office. The data is filtered to show only employees in the Marketing department. Below the table, a terminal window shows the SQL query used to retrieve the data: `select * from employees where department = 'Marketing' and office like '%East%';`. The terminal also shows the result of the query: 7 rows in set (0.001 sec).

employee_id	device_id	username	department	office
1000	a320b137c219	el Larson	Marketing	East-170
1052	a192b174c940	jd arrosa	Marketing	East-195
1075	x573y883z772	fb autist	Marketing	East-267
1088	k865l965m233	rg osh	Marketing	East-157
1103	NULL	randerss	Marketing	East-460
1156	a184b775c707	dellery	Marketing	East-417
1163	h679i515j339	cwilliam	Marketing	East-216

```
MariaDB [organization]> select *
-> from employees
-> where department = 'Marketing' and office like '%East%';

+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | el Larson | Marketing | East-170 |
| 1052 | a192b174c940 | jd arrosa | Marketing | East-195 |
| 1075 | x573y883z772 | fb autist | Marketing | East-267 |
| 1088 | k865l965m233 | rg osh | Marketing | East-157 |
| 1103 | NULL | randerss | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
+-----+-----+-----+-----+-----+

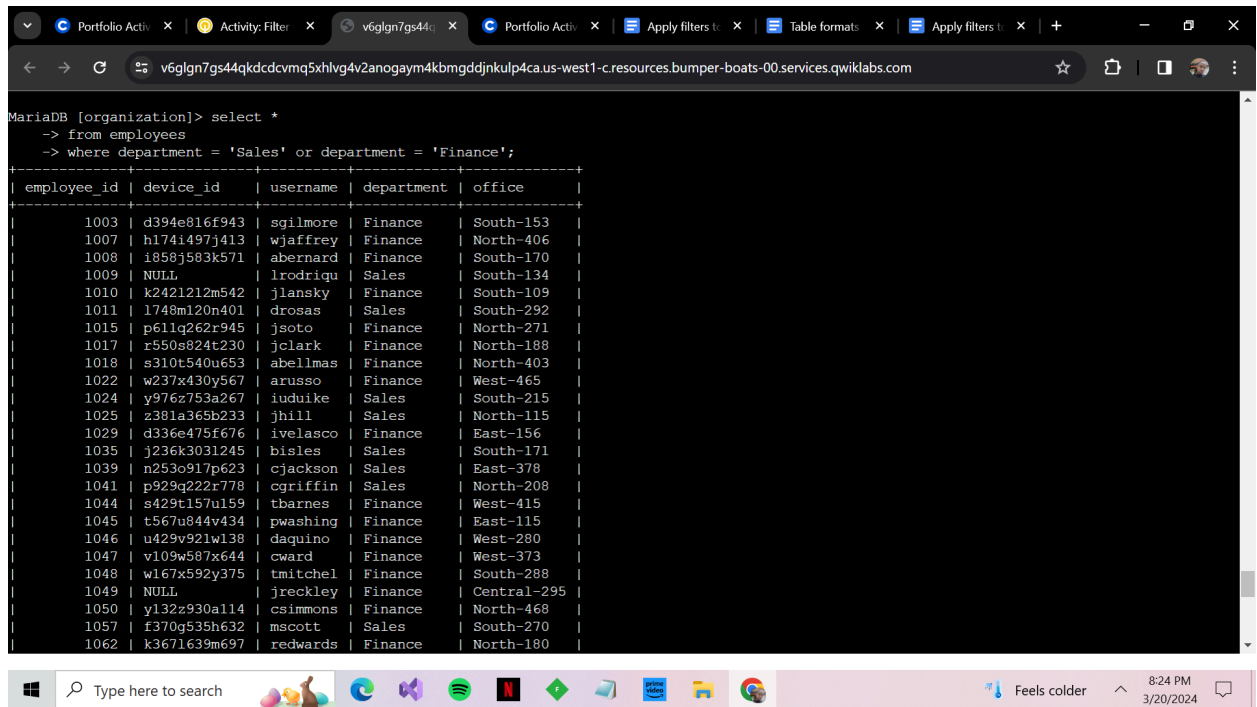
7 rows in set (0.001 sec)

MariaDB [organization]>
```

The screenshot shows how I retrieved all employees from marketing by using the where filter in SQL along with the like filter to query exactly what I want using the wildcard %%. The query now helps me see who needs updates.]

Retrieve employees in Finance or Sales

[



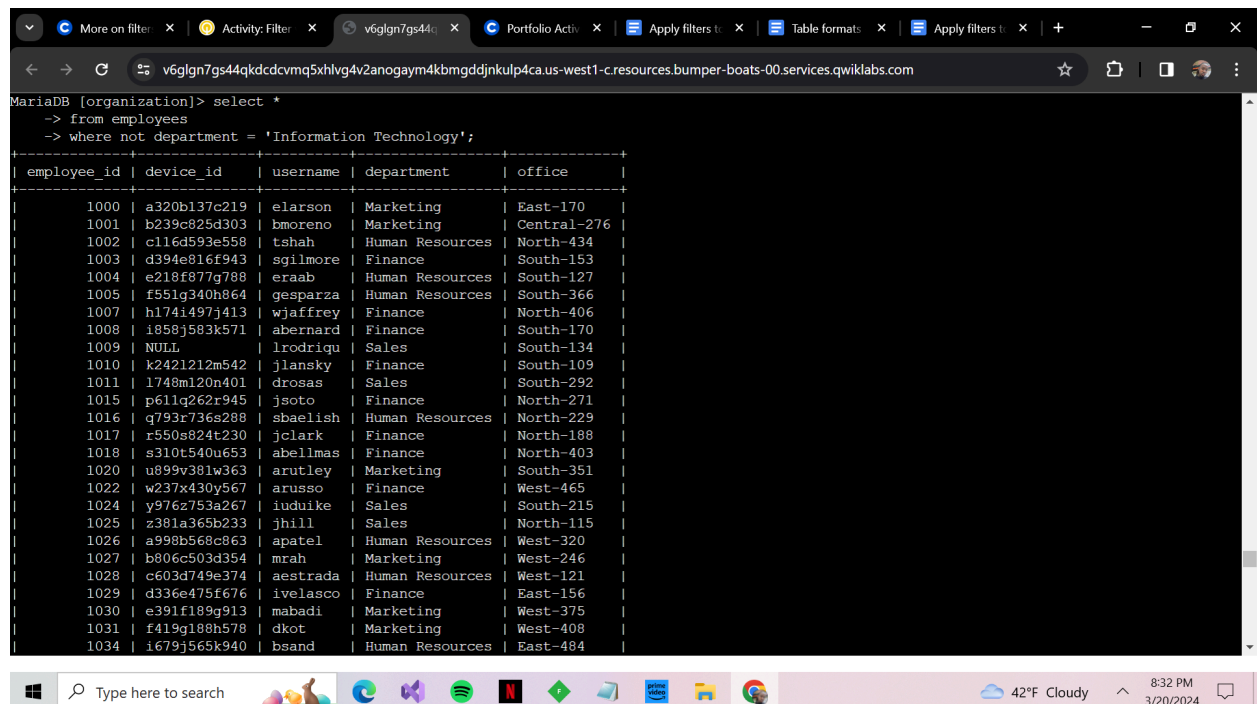
The screenshot shows a web browser window with a MariaDB interface. The query executed is: `select * from employees where department = 'Sales' or department = 'Finance';`. The result is a table with 5 columns: `employee_id`, `device_id`, `username`, `department`, and `office`. The table contains 26 rows of data, listing employees from both the Finance and Sales departments along with their device IDs and office locations.

employee_id	device_id	username	department	office
1003	d394e816f943	sgillmore	Finance	South-153
1007	h1741497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k2421212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasco	Finance	East-156
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1057	f370g535h632	mscott	Sales	South-270
1062	k367l639m697	redwards	Finance	North-180

The screenshot shows how I was able to filter sales along with the finance department by using the where and or filter in SQL. The query now is able to help me see who needs updates for their machines.]

Retrieve all employees not in IT

[



The screenshot shows a web browser window with a terminal interface for MariaDB. The terminal displays a SQL query and its results. The query is: `select * from employees where not department = 'Information Technology';`. The results are a table with 5 columns: `employee_id`, `device_id`, `username`, `department`, and `office`. The table contains 34 rows of data, listing employees from various departments such as Marketing, Human Resources, Finance, Sales, and IT. The browser's address bar shows the URL: `v6glgn7gs44qkdcvcmq5xhlgv4v2anogaym4kbgddjnkulp4ca.us-west1-c.resources.bumper-boats-00.services.qwiklabs.com`. The Windows taskbar is visible at the bottom of the screen.

```
MariaDB [organization]> select *
-> from employees
-> where not department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	el Larson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgillmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iudulke	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrah	Marketing	West-246
1028	c603d749e374	astrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375
1031	f419g188h578	dkot	Marketing	West-408
1034	i679j565k940	bsand	Human Resources	East-484

The screenshot shows how I was able to use another not filter along with where in order to get every department that was not a part of the IT department to filter out. The query now helps me see every employee except the IT department that needs an update for their machines.]

Summary

[In conclusion this is how I used my SQL filtering skills to tackle similar alerts and incidents that are in the cybersecurity field. The and, where, like, not, and lastly or are very powerful in helping me write less SQL queries and take what could have been two queries or more and turn it into one.]